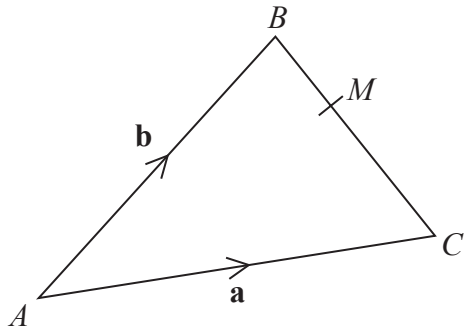
16



M is the point such that $BM : MC$ is $1 : 2$

Here is Charlie's method to find $\overrightarrow{BM}$ in terms of **a** and **b**.

$$\overrightarrow{BC} = \overrightarrow{BA} + \overrightarrow{AC}$$
$$= -\mathbf{b} + \mathbf{a}$$
$$= \mathbf{a} - \mathbf{b}$$
$$\overrightarrow{BM} = \frac{1}{2}\overrightarrow{BC}$$
$$= \frac{1}{2}(\mathbf{a} - \mathbf{b})$$

(a) Evaluate Charlie's method.

.....

.....

(1)

Martin expands $(2x + 1)(2x - 3)(3x + 2)$

He gets $12x^3 - 4x^2 - 17x + 6$

(b) Explain why Martin's solution cannot be correct.

.....

.....

(1)

(Total for Question 16 is 2 marks)